
Pose-Based Human Action Recognition for Athletic Movement Analysis

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<https://github.com/po8onthetrack/Deep-Learning-Project>

Introduction

This project is motivated by the author’s experience with badminton and bouldering, where reading how the body moves — a smash, a dynamic reach — is the first step toward better technique and fewer injuries. The goal is to recognise human movement from body pose alone: given a short sequence of 17 body-keypoint positions tracked across video frames from a single camera, classify it into one of the MPOSE2021 action categories, with no motion-capture equipment. Deep learning is appropriate because an action is defined by *how* the joints move over time, a spatio-temporal pattern that cannot be written as a fixed rule over keypoints; a recurrent network must learn both the pose configuration in each frame and its dynamics across frames. Reliable markerless action recognition is a building block for downstream movement analysis, training feedback, and injury-aware coaching.¹

Illustration

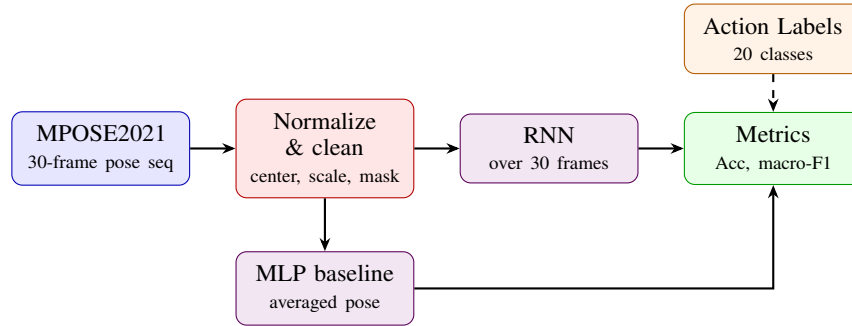


Figure 1: Action-recognition pipeline. MPOSE2021 provides 30-frame body-pose sequences (17 keypoints per frame) with human-annotated action labels. Each sequence is normalised (hip-centred, torso-scaled) and cleaned (low-confidence joints masked), then read by an RNN that models the temporal evolution of the pose; a static MLP on the time-averaged pose is the baseline. Both are scored against the ground-truth action labels (dashed). At inference, MoveNet converts raw video frames to keypoints (not shown).

Background & Related Work

BlazePose. Bazarevsky et al. (2020) showed that lightweight CNN pose estimation can run in real time on-device. This supports the choice of a frozen MoveNet-style backbone for fast keypoint extraction.

HRNet. Sun et al. (2019) maintains high-resolution feature maps for accurate keypoint localization. This motivates using an accurate but lightweight frozen pose estimator as the front-end that supplies keypoints to the action model.

ST-GCN. Yan et al. (2018) introduced spatial-temporal graph convolutions for skeleton-based action recognition, treating the body as a graph that evolves over time. This is the canonical deep approach to the task this project tackles.

¹Portions of the writing and code in this project were developed with the assistance of Claude (Anthropic, n.d.a) and Claude Code (Anthropic, n.d.b).

Hierarchical RNN. Du et al. (2015) showed that a recurrent neural network can recognise actions directly from sequences of skeleton joints, establishing the recurrent temporal approach this project builds on. It directly motivates the RNN at the core of our model.

Action Transformer. Mazzia et al. (2021) proposed a self-attention model for short-time pose-based action recognition and released MPOSE2021. This project uses the same dataset but starts from a simpler RNN as a strong, reproducible baseline.

Data Processing

The dataset is MPOSE2021 (Mazzia et al., 2021, ?; PIC4SeR, 2021), built for short-time pose-based action recognition: each sample is a 30-frame clip of 17 COCO body keypoints, shape $(30, 17, 3)$ as (x, y, conf) , with a human-annotated action label across 20 classes (e.g. walking, running, jumping, bending, boxing, hand-waving). Because the keypoints are pre-extracted — by PoseNet, in the same 17-keypoint COCO format that the MoveNet front-end produces at deployment — no pose estimation or hand-labelling is needed; the processing effort instead goes into making the raw keypoints model-ready through three steps: normalisation, cleaning, and a leakage-aware split.

First, each frame is *normalised* independently by centring the skeleton at the mid-hip — the midpoint of the left and right hip keypoints — and scaling by the torso length, taken as the mid-hip-to-mid-shoulder distance, so the input is invariant to body size, camera distance, and where in the frame the person stands, and the model sees pose geometry rather than raw image coordinates. Second, the keypoints are *cleaned*: joints with confidence below 0.2 — typically occluded or poorly detected — have their (x, y) position and confidence zeroed together, so the retained confidence channel doubles as a per-joint reliability flag and a failed detection is signalled to the network rather than entering it as a real joint position. Third, the clips are *split* 70 / 15 / 15% (train / validation / test) by stratified sampling on the action label to preserve class proportions, with the split kept subject-independent so that no actor seen in training reappears at validation or test time; the held-out test set is evaluated only once.

Architecture

The deployed system has two parts. MoveNet (TensorFlow Hub, 2024), a frozen pretrained CNN, supplies the 17 keypoints per frame at inference time (image $\rightarrow$ keypoints) and is never trained. The trained model is a recurrent neural network (RNN): each of the 30 frames contributes the normalised 51-dim pose vector (17×3) from data processing, and the RNN reads the frames in temporal order, passing its final hidden state to a fully connected layer that outputs a probability distribution over the action classes. An RNN is chosen because an action is characterised by how the pose changes over time; dropout and early stopping are used for regularisation, and weighted cross-entropy handles class imbalance.

Baseline Model

The baseline is a static classifier that discards temporal information: the 30 frames are averaged into a single 51-dim pose vector and classified by a small MLP — the same keypoint input, but with no sequence modelling. Comparing the RNN against this baseline isolates the value of temporal dynamics: if modelling how the pose evolves does not beat a single averaged pose, the added complexity is not justified. A majority-class predictor gives a trivial lower bound for additional context.

Ethical Considerations

MPOSE2021 contains pose data derived from human videos, so its licensing terms must be confirmed to permit academic use. The source clips may not represent all body types, ages, or ethnicities, so accuracy may be uneven across users. Because a predicted action label can drive downstream training feedback, a misclassification could mislead an athlete, so the system should be framed as a movement-analysis aid rather than an authoritative judgement of performance. Pose-based action recognition can also become surveillance, so any deployment that tracks real people must obtain informed consent.

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